Weidong Tang (汤维栋)

Chief Research Officer & Founding Member
Cardinal AI Lab, InfRec

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Biography

I am the Chief Research Officer and a founding member of Cardinal AI Lab, InfRec, leading a research team focused on advancing frontier AI research based on real-world long-horizon trajectory data. My research focuses on recursive self-improvement and long-horizon world interaction, enabling AI agents to accumulate long-term knowledge and autonomously control embodied robots.

I am also pursuing a B.E. in the School of Telecommunication Engineering at Xidian University, advised by Prof. Xiumei Wang. Previously, I was a research intern at Alibaba and TeleAI, and a Research Assistant at the HPC-AI Lab, National University of Singapore.

News

- 2026.07: I became a founding member of a startup.
- 2026.07: I completed my internship at Alibaba Group.
- 2026.04: Two papers accepted to ACL 2026.
- 2026.03: I completed my internship at TeleAI.
- 2026.02: One paper accepted to CVPR 2026.
- 2025.08: One paper accepted to PRCV 2025.
- 2025.07: One paper accepted to ICCV 2025.
- 2024.10: I joined the HPC-AI Lab at NUS as a Research Assistant.

Publications

* Equal contribution · † Corresponding author · ‡ Project leader

Core Publications

- The Trinity: Capability-Driven Recursive Self-Improvement for Multimodal Agents
  W. Tang[*], R. Wang[*], J. Zhang[*], H. Wang, H. Chen, Z. Zhang, Y. Ma, Z. Jiang, Z. Guo, H. Li, Z. Jiao, Z. Cui, X. Yin, S. Liang, Z. Wan, H. Lin, W. Zhao, K. Zhang, P. Zhou[†], O. Zhang[†], & Y. You[†].
  arXiv preprint, 2026 [Paper] [GitHub] [Project Page]
- GroupToM-Bench: Benchmarking Group Theory of Mind and Nonlinear Social Emergence in MLLMs
  W. Tang, J. Li, Y. Hou, Z. Mei, C. Zhang, X. Wan, Z. Liang, P. Zhou[†], Y. You, & W. Zhao[†].
  ACL 2026 (19% acceptance rate, CCF-A top-tier conference) [Paper] [Dataset]
- MagicBench: Diagnosing Visual Agency Loss and Semantic Dependency in Multimodal LLMs
  T. D. Huang, **W. Tang**[‡], W. Q. Xu, & X. Guo[†].
  ACL 2026 (19% acceptance rate, CCF-A top-tier conference)
- Efficient Video Object Segmentation and Tracking with Recurrent Dynamic Submodel
  W. Tang, Z. Liang, X. Wan, C. Zhu, Z. Xu, P. Zhou[†], Y. Song, Y. You, & W. Zhao[†].
  CVPR 2026 (25% acceptance rate, CCF-A top-tier conference) [Paper]

- Inference-Time Scaling for Visual AutoRegressive Modeling by Searching Representative Samples
  W. Tang, X. Wan, S. Li, & X. Wang[†].
  PRCV 2025 (29% acceptance rate, CCF-C) [Paper] [GitHub]

Collaborative Publications

- MDK12-Bench: A Multi-Discipline Benchmark for Evaluating Reasoning in Multimodal Large Language Models
  P. Zhou, X. Peng, F. Zhang, Z. Xu, J. Ai, Y. Qiu, W. Zhao, J. Song, C. Li, **W. Tang**, Z. Li, H. Zhang, Z. Li, X. Mao, Y. Feng, J. Sun, K. Wang, X. Chang, W. Shao, Y. You, & K. Zhang[†].
  AAAI 2026 (17% acceptance rate, CCF-A) [Paper] [GitHub]
- EA-ViT: Efficient Adaptation for Elastic Vision Transformer
  C. Zhu, W. Zhao[†], H. Zhang, Y. Zhou, **W. Tang**, S. Wang, Z. Yuan, Y. Shang, X. Peng, K. Wang, & D. Yang[†].
  ICCV 2025 (25% acceptance rate, CCF-A top-tier conference) [Paper] [GitHub]
- Pebench: A Fictitious Dataset to Benchmark Machine Unlearning for Multimodal Large Language Models
  Z. Xu, P. Zhou, **W. Tang**, J. Ai, W. Zhao, K. Wang, X. Peng, W. Shao, H. Yao, & K. Zhang[†].
  arXiv preprint, 2025 [Paper] [Dataset]
- IEBench: Benchmarking Unlearning Identity-Event Entangled Concepts for MLLM-Based Systems in Agentic Edge Networks
  Z. Xu, P. Zhou, **W. Tang**, J. Ai, W. Zhao, K. Wang, W. Shao, X. Peng, S. Zhao, H. Yao, & K. Zhang[†].
  IEEE Transactions on Network Science and Engineering, 2025 (IF=7.9, JCR-Q1) [Paper]

Research Experience

- **Cardinal AI Lab, InfRec** — Chief Research Officer (Founding Member) 07/2026 - Present
  - Leading a research team focused on advancing frontier AI research based on real-world long-horizon trajectory data.
  - Spearheading research on **Recursive Self-improvement (RSI)** and **Long-horizon World Interaction**, enabling AI agents to accumulate and use long-term knowledge while autonomously controlling embodied robots.
  - Driving research initiatives from problem formulation to prototype development, while collaborating with engineering teams for practical deployment.
  - Contributing to the growth of the lab, which recently secured nearly ten million RMB in funding to advance AI research.
- **Alibaba** — MLLM Research Intern 05/2026 - 07/2026
  - Researching scalable **post-training** techniques for multimodal foundation models, injecting e-commerce domain knowledge to improve product retrieval and find-similar-item applications.
  - Designed and implemented training and evaluation pipelines to improve **product representation learning**, **visual-semantic alignment**, and **retrieval performance** in large-scale e-commerce scenarios.
- **TeleAI** — LLM Research Intern 01/2026 - 03/2026
  - Worked on LLM training and inference **infrastructure for Huawei Ascend 910B** accelerators.
  - Focused on **numerical consistency** in both training and inference workflows.
- **HPC-AI Lab, National University of Singapore** — Research Assistant 10/2024 - 10/2025
  - Focusing on efficient machine learning and multimodal model evaluation.
  - Led end-to-end research projects from ideation to publication, resulting in three papers accepted by CVPR and ACL conferences. Recognized by Prof. Yang You as **one of the outstanding interns** in recent years.
  - Advised by Prof. Yang You, Prof. Wangbo Zhao, and Dr. Pengfei Zhou.

Education

- **Xidian University** 09/2023 - 07/2027 (expected)

  B.E., School of Telecommunication Engineering · GPA: 3.8/4.0
  Advised by Prof. Xiumei Wang.

Projects

- WowPage
  Weidong Tang, Yue Su.
  A clean and standalone homepage template. [GitHub] [RedNote]

  Developed an open-source, lightweight personal homepage framework featuring customizable layouts for researchers to present publications, projects, and academic profiles.

Professional Services

- AI Team Lead, Xidian-Inspur Club (06/2025 - Present).
- Reviewer for ICIC 2026, PRCV 2025-2026, EMNLP 2026, and ICLR 2026.

Selected Awards

- 2025 & 2026, Academic Scholarship, Xidian University (Top 5%).